



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO PONNI SERVICE STATION UNIT II

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

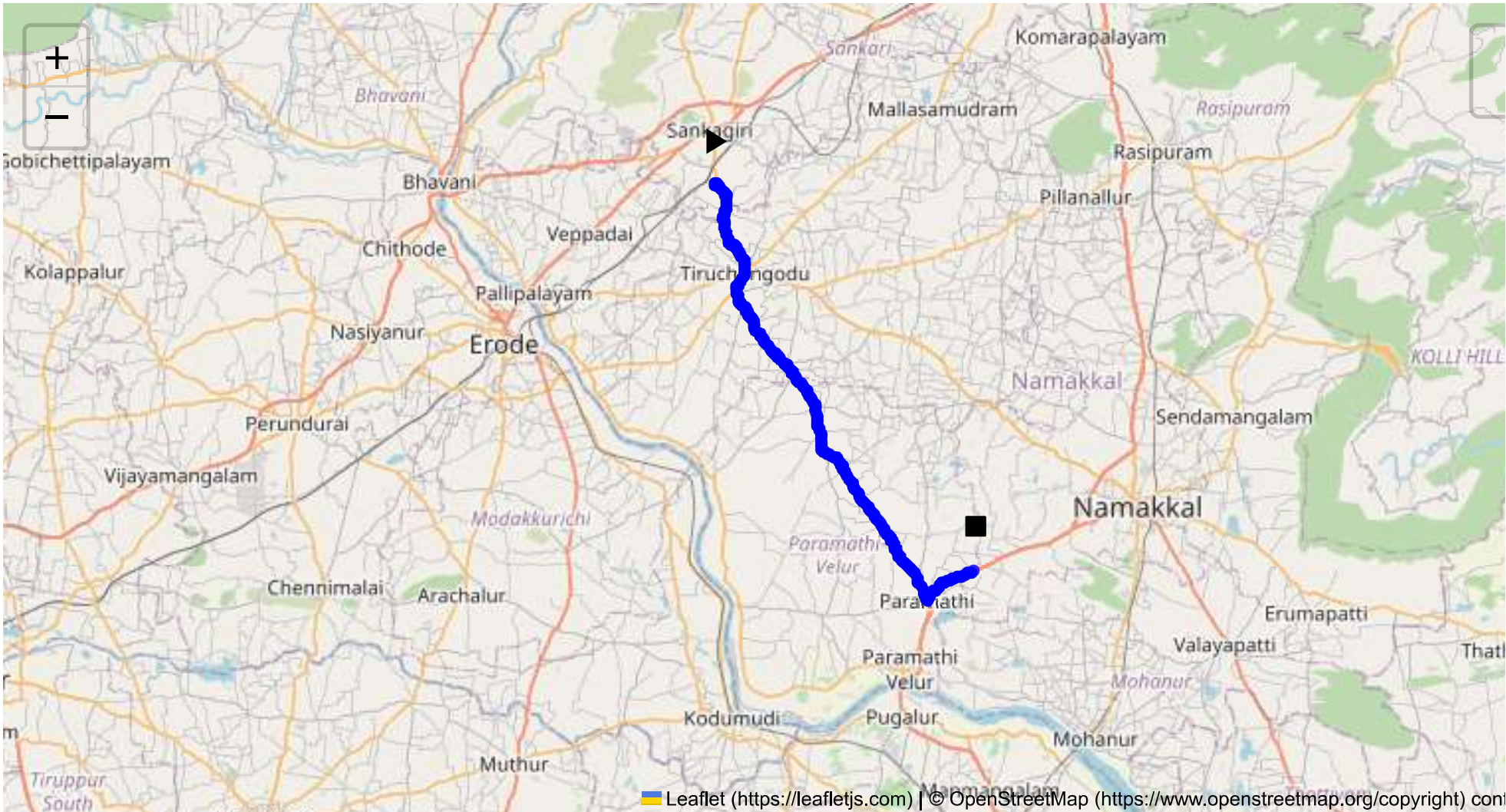
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 42.24 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.1 hours
Start: (11.4381, 77.8734)
End: (11.175631, 78.053579)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Namakkal traverses primarily via NH-544 and then NH-7. It is a stretch of approximately 42.24 kilometers that typically takes about 54 minutes to traverse in heavy vehicles carrying hazardous materials, with potential variances due to traffic and weather conditions.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather:** The region generally experiences tropical weather, with significant rainfall during the monsoon months (June to September). Summers can be extremely hot and dry.
- **Hazards:** Monsoon seasons can result in water-logged roads and decreased visibility. Flash flooding, especially in low-lying areas, could pose a danger.

3. Analysis of Traffic Patterns

- **Peak Hours:** Congestion is typically higher during the morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM) rush hours.
- **Congestion-Prone Areas:** Key congestion points can include urban intersections near Namakkal and areas around town centers where local traffic merges with highway traffic.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Generally, NH-544 and NH-7 are well-maintained national highways suitable for heavy vehicles. However, occasional potholes and uneven surfaces may be present, particularly after monsoon rains.
- **Infrastructure:** Adequate signage, but road lighting can be sparse in stretches, requiring caution during night travel.

5. Suggestions for Alternative Routes for Emergencies

Emergency detours can involve local roads parallel to NH-544 and NH-7. Consult updated local maps and real-time navigation tools during emergencies for the best alternate routes, particularly avoiding water-logged or crowded urban roads.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials requires compliance with safety guidelines by the Petroleum and Explosives Safety Organisation (PESO) and adherence to the Indian Motor Vehicles Act. Ensure all documentation and vehicle safety standards are rigorously maintained.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- Historical records indicate occasional accidents involving overloaded trucks, often due to inadequate braking systems or driver errors in high-speed zones.
- Incidents tend to occur near urban areas where abrupt merging of traffic happens.

8. Environmental Considerations and Sensitive Areas

Articulating passes near agricultural lands and small watershed zones; spilling hazardous materials could severely affect these environments. Awareness and caution are advised to prevent accidental contamination.

9. Analysis of Communication Coverage

- **Coverage:** Generally good mobile network coverage along this route, though minor dead zones might be experienced in rural stretches.
- **Action:** Equip drivers with multiple communication tools and check connectivity before and during the journey.

10. Estimated Emergency Response Times

- **Urban sections of Namakkal:** Approximately 15-20 minutes due to traffic.
- **Rural highways:** Can vary considerably, faster in clear conditions, potentially delayed during extreme weather.

11. Overall Summary of Risk Assessment

- **Moderate Risk:** Due to potential traffic congestion, weather impacts during monsoon, and requirement for adherence to local hazardous material transport regulations.
- **Recommendations:** Regularly check weather forecasts, plan travel to avoid peak congestion, ensure all safety and communication equipment is operational, and have a solid evacuation or detour plan in place.

By maintaining vigilance over these aspects, truck drivers can significantly mitigate risks while transporting hazardous materials on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Blind Spot	Blind Spot	11.15573, 78.02176	10 KM/Hr
8	Turn	Medium	11.16243, 78.02715	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	High	11.16241, 78.02729	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	college	PGP Group of Educational Institutions	11.1748266, 78.0472484	30 km/h	Medium
18	college	PGP College of Engineering	11.1748402, 78.0466235	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37868, 77.89498



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.15573, 78.02176



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16243, 78.02715



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.16241, 78.02729